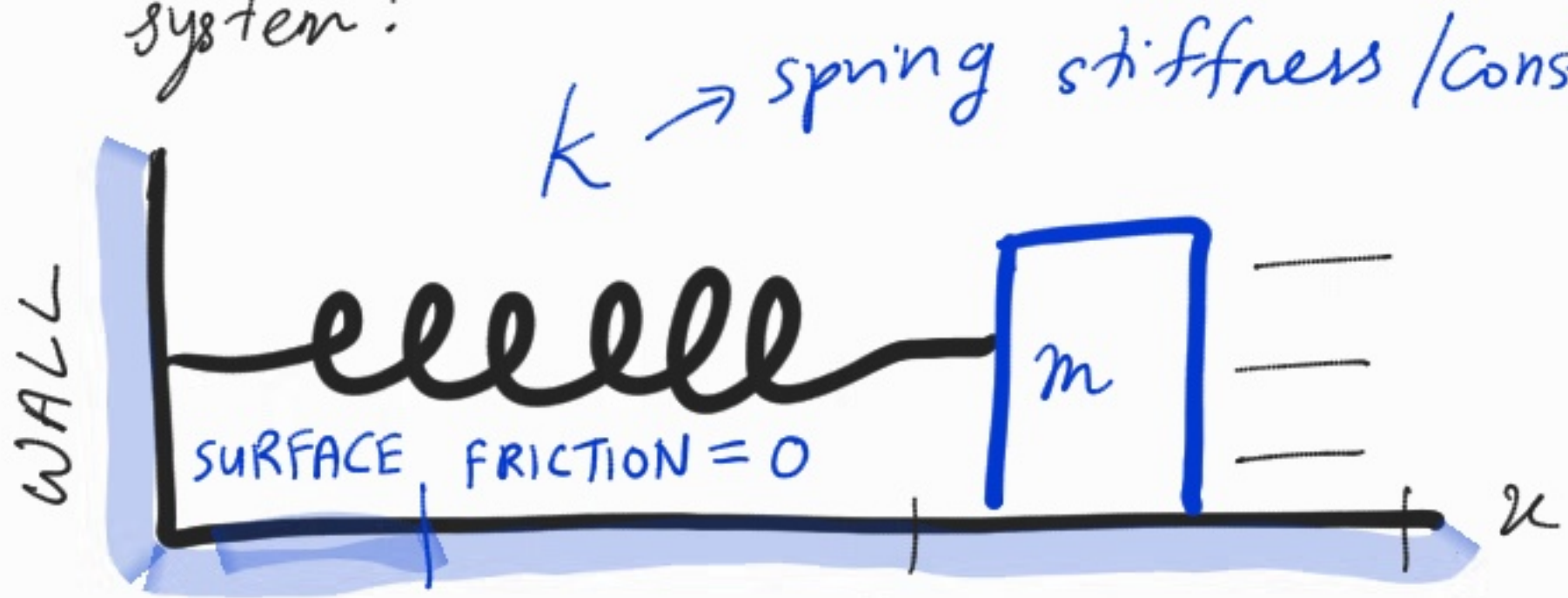
Hooke's Law.

- A mass-spring system is an example of simple harmonic motion.
- Consider this horizontal mass-spring system:



- The force is : $F = -kx$
- The spring oscillates between the two points.
- Let's determine the expression for the period & frequency of an oscillating mass-spring system.

Hooke's Law

- Newton's 2nd Law : $F = ma$
- Since $F = -kx$ for the mass-spring system :

$$-kx = ma = m\frac{d^2x}{dt^2} \quad \text{——(1)}$$

- we know that for simple harmonic motions :

$$x(t) = x_m \cos(\omega t + \phi)$$

$$a(t) = \frac{d^2x}{dt^2} = -x_m \omega^2 \cos(\omega t + \phi)$$

- Substituting in equation (1) we get :

$$-k(x_m \cos(\omega t + \phi)) = -m x_m \omega^2 \cos(\omega t + \phi)$$

Hooke's Law

$$-k[x_m \cos(\omega t+\phi)] = -m x_m \omega^2 \cos(\omega t+\phi)$$

$$\rightarrow \quad -k = -m\omega^2, \quad \therefore \quad \boxed{\omega = \sqrt{\frac{k}{m}}}$$

angular frequency

Since $\omega = \frac{2\pi}{T} = 2\pi f$,

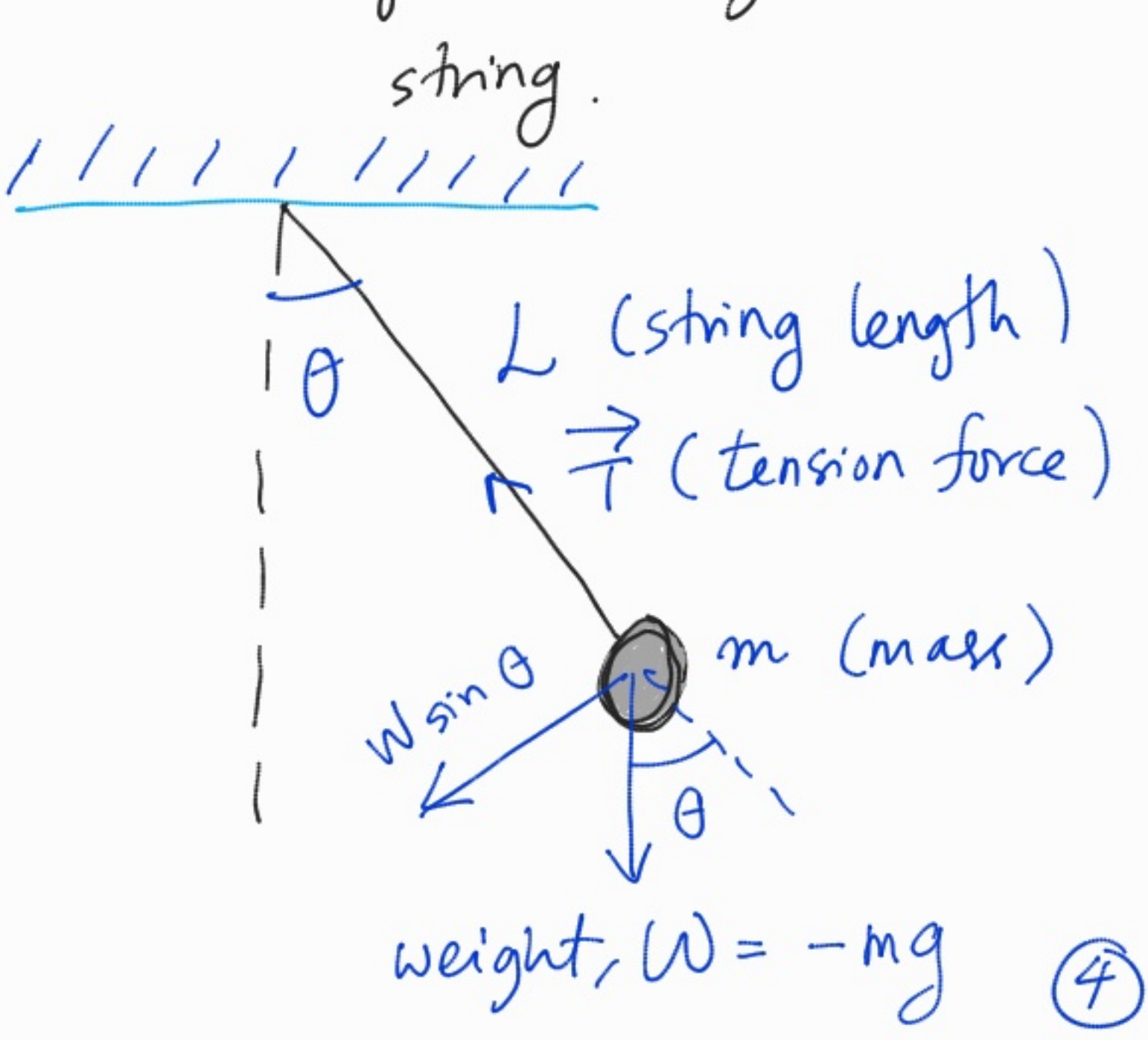
$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}} \quad \text{and}$$

$$\boxed{f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}}$$

Simple Pendulum

- Another interesting application of simple harmonic motion is the simple pendulum.

DEFINITION → A simple pendulum has an object with small mass suspended from a light wire or string.



Simple Pendulum

- Let's determine the expressions for period and frequency of a simple pendulum.

Using Newton's 2nd Law: $F = ma$

$W \sin\theta = ma$ ← tangential acceleration

Since $W = mg$,

$-\not{m}g \sin\theta = \not{m}a \rightarrow \boxed{a = -g\sin\theta}$

If θ is very small, then the following approximation can be applied:

$$\boxed{\sin\theta \approx \theta}$$

Simple Pendulum

- Then $a = -g \sin\theta$ becomes:

$$\boxed{a = -g\theta}$$

- Tangential acceleration, a can also be expressed in term of angular acceleration: $\alpha = d^2\theta / dt^2$

$$\boxed{a = L\alpha = L\frac{d^2\theta}{dt^2}}$$

- Combining both expressions:

$$-g\theta = L\frac{d^2\theta}{dt^2} \quad \text{or}$$

$$\boxed{L\frac{d^2\theta}{dt^2} + g\theta = 0}$$

Simple Pendulum

⇒ Recall Hooke's Law. The solution to the differential equation :

$$m\frac{d^2x}{dt^2} + kx = 0$$

$$x(t) = x_m \cos(\omega t + \phi)$$

$$\omega = \sqrt{\frac{k}{m}} \quad \left\{ T = \frac{1}{f} = 2\pi\sqrt{\frac{m}{k}} \right.$$

⇒ Therefore for the simple pendulum the same applies :

$$L\frac{d^2\theta}{dt^2} + g\theta = 0$$

$$\therefore \quad \omega = \sqrt{\frac{g}{L}} \quad \left\{ T = \frac{1}{f} = 2\pi\sqrt{\frac{L}{g}} \right.$$

Practice Problem 1

Determine the gravitational acceleration in a region if a simple pendulum has a period of 1.8 s and length of 76 cm. Compare this region's gravitational acceleration with earth's gravitational acceleration.

Solution.

For a simple pendulum, we found that : $T = 2\pi\sqrt{\frac{L}{g}}$

$$\therefore \quad g = \frac{4\pi^2 L}{T^2}$$

$$= \frac{4\pi^2 (0.76)}{(1.8s)^2}$$

$$= 9.26 \ ms^{-2}.$$

Comparing with earth : $g_0 = 9.81 \ ms^{-2}$

$$\delta = \left[\frac{9.81 - 9.26}{9.81}\right] \times 100\%$$

$$\approx 5.607\% \text{ weaker.}$$

Practice Problem 2

Bad shock absorbers cause cars to easily oscillate when going over bumps or after stopping. Determine the period & frequency of such an oscillation if an occupied car's total mass is 1 tonne and the force constant on the car's suspension system is 85 000 N/m.

Solution.

Consider the suspension system as a simple harmonic oscillator [analogous to a mass-spring system]

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} \quad \Big| \quad \begin{aligned} k &= 85000\ N/m \\ m &= 1000\ kg \end{aligned}$$

$$\therefore f = \frac{1}{2\pi}\sqrt{\frac{85000}{1000}} \approx 1.467\ Hz$$

$$T = \frac{1}{f} = 0.6815\ s$$

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